

END TERM EXAMINATION

FOURTH SEMESTER [BCA] JUNE 2024

Paper Code: BCAT -212

Subjects: Introduction to data Science

(Batch -2021 Onwards)

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No.1 which is Compulsory. Select one Question from each Unit.

Q1 Answer the Following

5x5=25

- (a) What are different Python Libraires?
- (b) Explain the concept of slicing array with suitable example.
- (c) What are different data security issues in data science?
- (d) How linear regression helps in data science?
- (e) How data frames are created from dictionary of series?

UNIT -I

- Q2 (a) What are different data pre-processing steps? [8.5]
(b) Explain different applications of data science. [4]
- Q3 (a) Explain different data collection strategies. [8.5]
(b) What are different data science roles? [4]

UNIT -II

- Q4 What is role of descriptive statistics in Data Science? Explain with suitable examples. [12.5]
- Q5 (a) Here are the runs scored by a Cricket team in a league of 12 matches – 110,120,100,150,110,140,130,180,120,250,140,110. Construct box plot for the same. [8.5]
(b) What is difference between skewness and kurtosis? [4]

UNIT -III

- Q6 (a) Write a code to implement different statistical function in Pandas. [6.5]
(b) What is pivoting? Write a Python code to implement pivot() and Pivot- table() on data frame. [6]
- Q7 (a) Explain the concept of broadcasting array in NumPy with suitable examples. [6]
(b) What are different data types available in NumPy? How negative indexing is implemented in Numpy? [6.5]

UNIT -IV

- Q8 What is a role of data science in investigating clinical data? Explain with suitable examples. [12.5]

- Q9 What is use of data science in forecasting demands? Explain with suitable examples. [12.5]

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FOURTH SEMESTER [BCA] JUNE- 2024

Paper Code: BCA-204

Subject: Software Engineering

Time: 3 Hours

Maximum Marks: 75

Note: Attempt all questions as directed. Internal Choice is indicated.

Q1. Write short notes on **any five** of the following: -

(5x5=25)

- (a) Differentiate between structural and functional testing.
- (b) Describe any two software size estimation techniques.
- (c) Discuss the need of data structure metrics.
- (d) What is software reverse engineering?
- (e) What are the characteristics of a good SRS?
- (f) Define cyclomatic complexity.
- (g) Discuss the RAD and Prototype SDLC models.

UNIT - I

Q2. Explain the various requirement elicitation techniques in detail. (12.5)

OR

Q3. Draw the E-R diagram, Use case diagram and Two Level DFD for Library Management System (12.5)

UNIT - II

Q4. Discuss the CoCoMo model in detail. Suppose that a project was estimated to be 600KLOC. Calculate the effort and development time for each of the three modes i.e., organic, semidetached and embedded. (12.5)

OR

Q5. Compare the Walston-Felix and SEL model for a S/W to be developed that involves an effort of 8 persons- years.

- (i) Calculate the number of line of source code that can be produced
- (ii) Calculate the duration of the development
- (iii) Calculate the productivity in LOC/PY (12.5)
- (iv) Calculate the average manning.

UNIT - III

Q6. What is a software module and cohesion? Discuss the various types of module cohesions that exists in various modules of a software. (12.5)

OR

Q7. Write short notes on

- (a) Total Quality Management(TQM)
- (b) Configuration Management
- (c) Software Re-Engineering

UNIT- IV

Q8. Consider the problem of identification of types of a triangle on the basis of the three sides in the domain (1,100). Generate the total number of test cases for performing Equivalence Class Testing and Decision table based testing for the said problem. (12.5)

OR

Q9. Discuss the need of software maintenance and also the different types of software maintenance performed to ensure the appropriate working of a software. (12.5)

END TERM EXAMINATION

FOURTH SEMESTER [BCA] JUNE 2024

Paper Code: BCA-222

Subjects: Digital Marketing

Time: 3 Hours

Maximum Marks: 75

Note: Attempt all questions as directed. Internal Choice is indicated.

Q1. Attempt **any five** of the following: --

(5x5=25)

- a) Differentiate digital marketing and traditional marketing?
- b) Explain Internet Marketing and give its advantages?
- c) Describe the role of Influencer marketing in social media marketing?
- d) Differentiate Inbound and outbound marketing?
- e) State the meaning of content marketing with suitable examples.
- f) Explain 4 C's of Digital Marketing.
- g) What do you mean by campaign criteria?
- h) Define SMM tools like Hootsuite, Canva etc.?

UNIT- I

Q2. Explain Digital Marketing channels in Detail. What are business Models? Explain its types also.

(12.5)

OR

Q3. Explain in detail the POEM framework in Digital Marketing strategy. Explain its benefits and Challenges?

(12.5)

UNIT- II

Q4. Explain social media marketing strategies and social media platforms in detail which can help in marketing a new venture.

(12.5)

OR

Q5. Discuss the role and importance of Influencer and Content marketing in online branding. Explain with the help of an example

(12.5)

UNIT- III

Q6. Explain role of SEO keywords in SEO? Explain the Organic Vs Non Organic SEO in details.

(12.5)

OR

Q7. Explain **any Two** the following:

(12.5)

- a) Google Adwords
- b) Blog marketing
- c) Paid marketing

UNIT- IV

Q8. Explain the process of Web analytics? How google analytics works? Why is web Analytics is important for company performance?

(12.5)

OR

Q9. Write short note on with their uses:

(12.5)

- a) Explain Hot Jar
- b) Behavior analytics

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FOURTH SEMESTER [BCA] JUNE- 2024

aper Code: BCA-202

Subject: Java Programming

ime: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.no. 1 which is compulsory. Select one question from each unit.

Q1 Answer the following:

(2.5 x 10=25)

- (a) What are the key differences between Java and C++?
- (b) Explain different types of access specifiers available in Java?
- (c) Explain the features of Java.
- (d) How Garbage Collection is achieved in Java?
- (e) State the purpose of Static block.
- (f) Explain checked and unchecked exceptions.
- (g) Explain Adapter Class.
- (h) What is the use of Super and final keyword?
- (i) Differentiate abstract class and interface.
- (j) What is the difference between int and integer.

Unit -I

- Q2 (a) What is a constructor? Write a program to show constructor overloading. (6)
(b) Write a program to show 5 string methods. What is a StringBuffer class? (6.5)

- Q3 Explain Java architecture in brief. What are the types of applications in Java? (12.5)

Unit -II

- Q4 (a) What is inheritance? Write a program to show multilevel inheritance. (6)
(b) Explain the usage of interface in Java? Give an example of a class which implements interface. (6.5)
- Q5 (a) What is package? How to use two different packages in a Java class? Give example. (6)
(b) Explain exception handling with suitable example. Differentiate between throw and throws? (6.5)

Unit -III

- Q6 (a) What are the byte and character streams in file handling? (6)
(b) Write a program to read and write the content of a file in text mode. (6.5)
- Q7 Explain the life cycle of Thread and methods to implement thread with suitable code. (12.5)

Unit -IV

- Q8 (a) Create a login form using Swing Controls. (6)
(b) What are the different types of Layout Managers? (6.5)
- Q9 Explain the JDBC Architecture. What is JDBC Driver and Connection statement? (12.5)

END TERM EXAMINATION

FOURTH SEMESTER [BCA] JUNE 2024

Paper Code: BCA-206

Subject: Introduction to Management & Entrepreneurship Development

Time: 3 Hours

Maximum Marks: 75

Note: Attempt all questions as directed. Internal choice is indicated.

- Q1 Attempt any Five of the following questions (5x5=25)
- a) Define entrepreneurship with an example.
 - b) State any five skills required to be an entrepreneur.
 - c) Name the steps of creativity in entrepreneurship and elaborate.
 - d) Briefly discuss the stages of innovation in entrepreneurship.
 - e) State the objectives of design thinking in entrepreneurship.
 - f) Justify the statement "An entrepreneur can be considered a role model".
 - g) Define the Social Responsibility of business with an example.
 - h) Define the main components of a comprehensive business plan.

- Q2 a) State any six significance of various public agencies in entrepreneurship development. (6)
b) State any seven significance of various private agencies in entrepreneurship development. (6.5)

OR

- Q3 a) State any seven roles to be played by entrepreneurs in the development of a country's economy. (6.5)
b) Enumerate any six personality traits that are significant in an entrepreneur to be successful. (6)

- Q4 a) How do creativity and innovation contribute towards entrepreneurship? (8)
b) What are the day-to-day operational activities done in entrepreneurial business? (4.5)

OR

- Q5 a) Define innovation in entrepreneurship. (2.5)
b) Elaborate any ten examples of innovative decision-making toward entrepreneurial problem-solving from different sectors. (10)

- a) Elaborate on the role of changing mindset and out-of-the-box thinking in entrepreneurship in present-day business. (8)
b) As a mentor, how can an entrepreneur contribute to entrepreneurship development? (4.5)

OR

- a) How the global Indian entrepreneurs have achieved success across the globe? (6)
b) The modern entrepreneurs have achieved success in their respective sectors, justify. (6.5)

How is management different from leadership? (5)
How do the roles of entrepreneurs and managers differ within an organization? (7.5)

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OR

- Q9 a) A well-developed business plan helps entrepreneurs communicate their vision and strategy to stakeholders like investors and partners, explain. (5)
b) A business plan becomes a roadmap for entrepreneurs at various stages of starting and managing a venture. Justify this statement. (7.5)

BCA-206

P_{1/2}

BCA-206

P_{2/2}